

Diet and Lifestyle: A Statistical Examination of Their Ties to Obesity

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I. INTRODUCTION

Obesity is a pervasive global health challenge, with profound implications for public health and well-being. In Colombia, Peru, and Mexico, the prevalence of obesity has risen significantly in recent years. According to the Global Obesity Observatory, the obesity rate in Mexico has increased significantly from 19.4% in 2000 to a staggering 31.8%. What's even more concerning is that when combined with overweight individuals, the total surpasses 70% [1]. This data underscores the severity of the obesity issue in Mexico and provides a compelling backdrop for our research.

Our specialized research is focused on estimating obesity levels in individuals from Colombia, Peru, and Mexico. We hypothesize that there is one significant relationship between obesity and their eating habits and one significant relationship between obesity and their lifestyle habits.

Our primary aim is to provide a thorough assessment of obesity levels and the factors contributing to obesity in these countries, with a particular focus on understanding the interplay between their eating habits and obesity, as well as lifestyle habits and obesity. We will employ a combination of descriptive analysis, data set modeling, and inference generation to achieve this goal. This research will shed light on the specific factors and patterns contributing to obesity in Colombia, Peru, and Mexico.

II. METHODS & MATERIALS

A. Data Set

The data set is designed for estimating obesity levels in individuals from Mexico, Peru, and Colombia, offering valuable insights into a diverse range of eating habits and lifestyle habits. It encompasses 2111 observations with 17 variables, collected through a web-based survey accessible for 30 days. These attributes cover both eating habits (e.g., frequent consumption of high-caloric food, vegetable consumption frequency) and

physical conditions (e.g., calories consumption monitoring, physical activity frequency). The data set also includes demographic variables such as *Gender*, *Age*, *Height*, and *Weight*. Each observation is thoughtfully labeled with the class variable *NObeyesdad* (Obesity Level), allowing us to classify individuals into different categories, ranging from "Insufficient Weight" to "Obesity Type III". Notably, 77% of the data has been synthetically generated using the tool Weka and the filter SMOTE (Synthetic Minority Over-sampling Technique), while the remaining 23% has been collected directly from users through a web platform.

These abbreviations are pertinent to our research article, serving as key terminology in our data and analysis, each accompanied by its respective explanation:

Table 1: Abbreviations used in the data

FAVC	Frequent Consumption of High Caloric Food
FCVC	Frequency of Consumption of Vegetables
NCP	Number of Main Meals
CAEC	Consumption of Food between Meals
SMOKE	Smoking Status
CH2O	Consumption of Water Daily
CALC	Consumption of Alcohol
NObeyesdad	Obesity Result
SCC	Calories Consumption Monitoring
FAF	Physical Activity Frequency
MTRANS	Transportation Used
TUE	Time Using Technology Devices

Our criteria for assessing BMI are based on the formula: $BMI = \text{weight} / (\text{height} * \text{height})$, with the following classifications:

Normal: BMI falls within the range of 18.5 to 24.9.

Overweight: BMI falls within the range of 25.0 to 29.9.

Obesity I: BMI falls within the range of 30.0 to 34.9.

Obesity II: BMI falls within the range of 35.0 to 39.9.

Obesity III: BMI is higher than 40.

The data can be used for a wide range of data mining tasks, including classification, prediction, segmentation, and association, and it is available in both CSV and ARFF formats for compatibility with

data analysis tools like Weka. This comprehensive data set is particularly valuable for research and the development of computational tools related to obesity level estimation.

B. Study Population

We had two sets of raw data, one concerning eating habits relationship with obesity, and the other related to the lifestyle habits relationship with obesity. Some of the original data contained many decimal places that seemed to be irrelevant, possibly due to data quality issues. Before processing the data, we edited it to have an appropriate number of decimal places.

In SAS, we merged two data sets and combined the two data sets. Subsequently, we employed the “proc means” procedure to generate descriptive statistics for each measurement in the data set.

Table 2: Descriptive statistics of each measurement

Variable	N	Mean	Std Dev	Median	Minimum	Maximum
Age	2111	24.31	6.36	23.0	14.00	61.00
Height	2111	1.70	0.09	1.7	1.45	1.98
Weight	2111	86.59	26.19	83.0	39.00	173.00
FCVC	2111	2.42	0.58	2.0	1.00	3.00
NCP	2111	2.68	0.81	3.0	1.00	4.00
CH2O	2111	2.01	0.69	2.0	1.00	3.00
FAF	2111	1.01	0.90	1.0	0	3.00
TUE	2111	0.66	0.67	1.0	0	2.00

The histogram in Figure 1 provides a straightforward representation of the number of individuals in each obesity level category. It was evident that the number of individuals in each category was relatively evenly distributed in the graphical representation which is suitable for further analysis.

In Figure 2, the pie chart illustrates the relationship between obesity and a family history of overweight. Among individuals with a family history of overweight, 83.72% of them exhibit varying degrees of obesity or overweight. In contrast, among individuals without a family history of overweight, only 25.71% are overweight, and none are obese. This indicates that a family history of overweight has a significant impact on an individual's weight status.

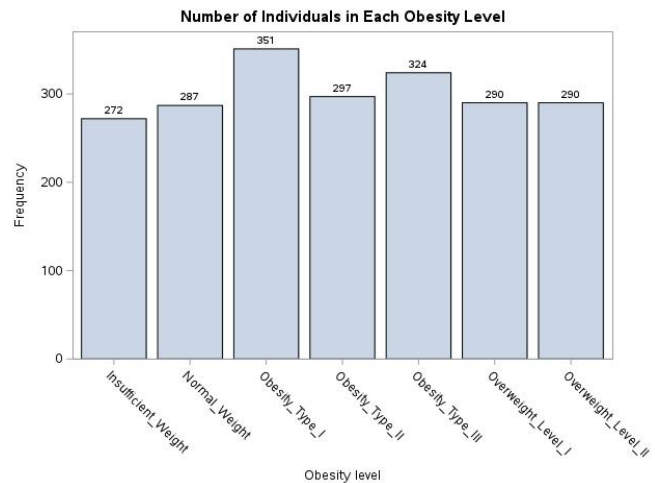


Figure 1: Histogram of Obesity Level Data

C. Statistical Methods

1. Chi-Square Analysis

We used Chi-Square Analysis in SAS to separately explore the significance of relationships between obesity and eating habits, and obesity and lifestyle habits, respectively.

2. Random Forest

We used a random forest model to perform multiclass classification and weight analysis on this data set based on different eating habits and lifestyles, exploring how these habits affect obesity levels.

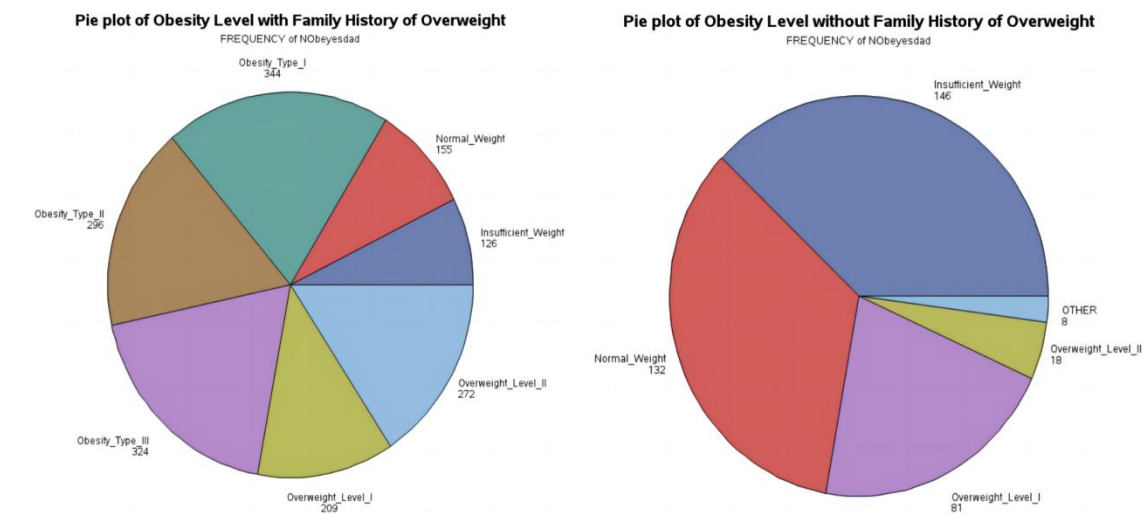


Figure 2: Family History and Weight Status

III. RESULTS

1. Chi-Square Analysis

In the following two Chi-Square analyses, we have set *NObeyesdad* as the dependent variable and *NObeyesdad* = "Normal_Weight" as the reference category.

1.1 Obesity vs. Eating Habits

We explored the relationship between the dependent variable and the independent variables: *FAVC*, *FCVC*, *NCP*, *CAEC*, *SMOKE*, *CH2O* and *CALC*. From Table 3, we can discover that all seven independent variables have statistical significance through the Wald Chi-Square and $Pr > ChiSq$, and the relationships between them and the dependent variable are all quite significant. Among them, the effect of *CAEC* is the most significant, 315.3006, while the effect of *SMOKE* is the least significant, 23.7193. We can also check R-Square from Table 3 and the value is 0.6255, which means approximately 62.55% of the dependent variable can be explained by the model, and the model has a relatively satisfactory level of fit.

1.2 Obesity vs. Lifestyle Habits

We explored the relationship between the dependent variable and the independent variables: *SCC*, *FAF*, *MTRANS* and *TUE*. Similar to 1.1, we can check Wald Chi-Square, $Pr > ChiSq$ and R-Square values from Table 3. We can discover that all four independent variables have statistical significance, and the relationships between them and the dependent variable are all relatively significant. Among them, the effect of *MTRANS* is the most significant, 106.2450, while the effect of *TUE* is the least significant, 40.6623. We can also check R-Square from Table 3 and the value is 0.2419, which means only approximately 24.19% of the dependent variable can be explained by the model, and it appears that the model exhibits a suboptimal fit.

Table 3: Chi-Square Analysis

Obesity vs. Eating Habits			Obesity vs. Lifestyle Habits		
Variable	Wald Chi-Square	Pr > ChiSq	Variable	Wald Chi-Square	Pr > ChiSq
FAVC	119.9518	<.0001	SCC	45.2272	<.0001
FCVC	71.4981	<.0001	FAF	79.8576	<.0001
NCP	105.6870	<.0001	MTRANS	106.2450	<.0001
CAEC	315.3006	<.0001	TUE	40.6623	<.0001
SMOKE	23.7193	0.0006			
CH2O	52.5396	<.0001			
CALC	124.4447	<.0001			
Inspection Standard	R-Square	0.6255	Inspection Standard	R-Square	0.2419

2. Random Forest

We used the obesity level *NObeyesdad* as the target variable. For both eating habits and lifestyle habits, a random forest with 250 decision trees is implemented. Each decision tree has a max depth of 20. Table 4 shows the loss reduction and variable importance analysis for both random forests.

2.1 Obesity vs. Eating Habits

The random forest model based on eating habits achieved an accuracy of 58.6%. For variable importance analysis, according to the Gini and OOB-Gini from Table 4, experiment results basically align with the Chi-Square analysis. *CAEC*, *FCVC*, and *NCP* variables possess the top three highest Gini values, highlighting their importance in the model. *SMOKE*, with the lowest Gini, OOB Gini, and Number of Rules, is the least impactful among those presented.

2.2 Obesity vs. Lifestyle Habits

The random forest model based on lifestyle habits is not as good as the previous eating habits model, with an unsatisfied accuracy of 36.7%. The Gini and OOB-Gini for this model indicate that the variable importance comes in order of *FAF*, *TUE*, *MTRANS* and *SCC*, which also differs from previous Chi-Square analysis.

Table 4: Variable Importance and Relationship Analysis of Random Forest

Models	Variable	Number of Rules	GINI	OOB GINI	Margin	OOB Margin
Eating Habits	CAEC	1447	0.052445	0.045330	0.02093	0.01475
	FCVC	3170	0.057190	0.043622	-0.00357	-0.01547
	NCP	5613	0.059337	0.036312	0.03847	0.01883
	CALC	1768	0.038967	0.029602	0.01481	0.00760
	FAVC	1075	0.017848	0.012329	0.00564	0.00139
	CH2O	5955	0.038409	0.011097	0.04144	0.01890
	SMOKE	187	0.001724	0.000498	0.00220	0.00114
Lifestyles	FAF	2724	0.030876	0.017284	-0.00569	-0.01584
	TUE	1785	0.023047	0.014146	-0.01285	-0.01941
	MTRANS	1051	0.018374	0.012317	-0.01485	-0.01998
	SCC	792	0.012746	0.008928	0.00364	0.00068

IV. CONCLUSIONS

The model evaluating obesity against eating habits demonstrates satisfactory accuracy. Additionally, the variable importance derived from the random forest aligns well with the Chi-Square analysis, confirming the reliability and consistency of the findings from both methods.

In contrast, the model relating obesity to lifestyle habits shows limited accuracy. Interestingly, there is a discrepancy between the random forest's variable importance and the Chi-Square analysis. Such variations imply that while certain lifestyle factors might statistically correlate with obesity, they might not be as predictive when considering multiple variables in tandem.

From the overall study, it is evident that the influence of eating habits on obesity overshadows that of lifestyle patterns. Particularly, variables like *CAEC* have showcased a strong positive correlation with obesity, making them pivotal in any obesity-related discussions. On the contrary, the variable *SMOKE*, despite its statistical significance, demonstrated minimal impact, hinting at either a weak positive or a potential negative correlation with obesity.

V. DISCUSSION

Obesity has been associated with biological, behavioral and environmental determinants. Numerous research studies have illustrated the role lifestyle factors (physical inactivity, poor diet, insufficient sleep, and excessive sedentary behavior) play in obesity. Physical activities are considered to be one of the greatest contributors to obesity and the health risks associated with it [2]. Our results also align with those reported in Saudi Arabia [3] and Portugal [4] which found a link between the consumption of high-calorie foods and general obesity [5,6,7]. Our study observed a subtle relationship between obesity and TUE (screen time), a finding that aligns with several other research conclusions [3,8,9]. Contrarily, some studies have highlighted a distinct link between adiposity and prolonged screen exposure [10,11]. A potential underlying factor could be the intake of unhealthy

foods, particularly sugar-laden snacks, during screen activities.

The present research has several constraints. As the data stems from a cross-sectional investigation, it precludes establishing causative links between obesity and the studied variables. Relying on self-reported tools for lifestyle indicators might introduce inaccuracies. Additionally, the food frequency survey omit details on portion sizes. On the upside, the study benefits from a substantial sample and employs a verified questionnaire. While our models demonstrate varying degrees of success in predicting obesity based on eating and lifestyle habits, the relatively lower accuracy of the lifestyle habits model suggests there are other unexplored variables or external factors that play crucial roles in obesity.

As future work, it would be beneficial to validate our findings with larger, more diverse, and purely organic data sets. Expanding the range of variables and incorporating qualitative data, like personal narratives or regional dietary influences, might offer richer insights. Moreover, advancements in computational tools and machine learning techniques can be harnessed to refine our analyses and provide a more nuanced understanding of obesity determinants.

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REFERENCES

- [1] Estimation of obesity levels based on eating habits and physical condition . (2019). UCI Machine Learning Repository.
<https://doi.org/10.24432/C5H31Z>.

- [2] Katzmarzyk, Peter T., et al. "Physical activity, sedentary time, and obesity in an international sample of children." *Med Sci Sports Exerc* 47.10 (2015): 2062-9.
- [3] Al-Hazzaa, Hazzaa M., et al. "Lifestyle factors associated with overweight and obesity among Saudi adolescents." *BMC public health* 12.1 (2012): 1-11.
- [4] Martinez-Gomez, David, et al. "Combined influence of lifestyle risk factors on body fat in Spanish adolescents—the Avena study." *Obesity facts* 4.2 (2011): 105-111.
- [5] Al-Haifi, Ahmad R., et al. "Relative contribution of physical activity, sedentary behaviors, and dietary habits to the prevalence of obesity among Kuwaiti adolescents." *Food and nutrition bulletin* 34.1 (2013): 6-13.
- [6] Musaiger, Abdulrahman O., Fatima O. Nabag, and Mariam Al-Mannai. "Obesity, dietary habits, and sedentary behaviors among adolescents in Sudan: alarming risk factors for chronic diseases in a poor country." *Food and nutrition bulletin* 37.1 (2016): 65-72.
- [7] Borges, Camila Aparecida, et al. "Dietary patterns associated with overweight among Brazilian adolescents." *Appetite* 123 (2018): 402-409.
- [8] dos Santos Cavalcanti, Cláudio Barnabé, et al. "Abdominal obesity in adolescents: prevalence and association with physical activity and eating habits." *Arquivos Brasileiros de Cardiologia* 94.3 (2010): 350-356.
- [9] Porslow, Lisa R., et al. "Differences in physical activity and sedentary time in relation to weight in 8–9 year old children." *International Journal of Behavioral Nutrition and Physical Activity* 5 (2008): 1-9.
- [10] Carson, Valerie, et al. "Systematic review of sedentary behaviour and health indicators in school-aged children and youth: an update." *Applied physiology, nutrition, and metabolism* 41.6 (2016): S240-S265.
- [11] Chinapaw, Mai, Teatske Altenburg, and Johannes Brug. "Sedentary behaviour and health in children—Evaluating the evidence." *Preventive Medicine* 70 (2015): 1-2.

```

1 *input eating habits raw data*;
2 libname GROUP "/home/u63615521/";
3 run;
4
5 PROC IMPORT OUT= GROUP.TB1
6 DATAFILE= "/home/u63615521/Biostat203A/midterm/eating_habits_edited data.csv"
7 DBMS=CSV REPLACE;
8 GETNAMES=YES;
9 DATAROW=2;
10 RUN;
11
12 data GROUP.TB1 (label="Eating Habits");
13 infile "/home/u63615521/Biostat203A/midterm/eating_habits_edited data.csv" DSD firstobs=2;
14 informat Gender $6.;
15 informat Age BEST12.;
16 informat Height BEST12.;
17 informat Weight BEST12.;
18 informat family_history_with_overweight $3.;
19 informat FAVC $5.;
20 informat FCVC BEST12.;
21 informat NCP BEST12.;
22 informat CAEC $12.;
23 informat SMOKE $5.;
24 informat CH2O BEST12.;
25 informat CALC $12.;
26 informat NObeyesdad $19.;
27
28 format Gender $6.;
29 format Age BEST12.;
30 format Height BEST12.;
31 format Weight BEST12.;
32 format family_history_with_overweight $3.;
33 format FAVC $5.;
34 format FCVC BEST12.;
35 format NCP BEST12.;
36 format CAEC $12.;
37 format SMOKE $5.;
38 format CH2O BEST12.;
39 format CALC $12.;
40 format NObeyesdad $19.;
41
42 input
43 Gender $
44 Age
45 Height
46 Weight
47 family_history_with_overweight $
48 FAVC $
49 FCVC
50 NCP
51 CAEC $
52 SMOKE $
53 CH2O
54 CALC $
55 NObeyesdad $;
56
57 label Gender="Gender"
58 Age="Age"
59 Height="Height"
60 Weight="Weight"
61 family_history_with_overweight="Family History with Overweight"
62 FAVC="Frequent Consumption of High Caloric Food"
63 FCVC="Frequency of Consumption of Vegetables"
64 NCP="Number of Main Meals"
65 CAEC="Consumption of Food between Meals"
66 SMOKE="Smoking Status"
67 CH2O="Consumption of Water Daily"
68 CALC="Consumption of Alcohol"
69 NObeyesdad="Obesity Result";
70
71 *input lifestyle habits raw data*;
72 libname GROUP2 "/home/u63615521/";
73 run;
74
75 PROC IMPORT OUT= GROUP.TB2
76 DATAFILE= "/home/u63615521/Biostat203A/midterm/lifestyle_habits_edited data.csv"
77 DBMS=CSV REPLACE;
78 GETNAMES=YES;
79 DATAROW=2;
80 RUN;
81
82 data GROUP.TB2 (label="Physical Condition");
83 infile "/home/u63615521/Biostat203A/midterm/lifestyle_habits_edited data.csv" DSD firstobs=2;

```



```

82 informat Gender $6.;
83 informat Age BEST12.;
84 informat Height BEST12.;
85 informat Weight BEST12.;
86 informat family_history_with_overweight $3.;
87 informat SCC $3.;
88 informat FAF BEST12.;
89 informat MTRANS $21.;
90 informat TUE BEST12.;
91 informat NObeyesdad $19.;
92 format Gender $6.;
93 format Age BEST12.;
94 format Height BEST12.;
95 format Weight BEST12.;
96 format family_history_with_overweight $3.;
97 format SCC $3.;
98 format FAF BEST12.;
99 format MTRANS $21.;
100 format TUE BEST12.;
101 format NObeyesdad $19.;
102
103 input Gender $
104 Age
105 Height
106 Weight
107 family_history_with_overweight $
108 SCC $
109 FAF
110 MTRANS $
111 TUE
112 NObeyesdad $;
113
114 label Gender="Gender"
115 Age="Age"
116 Height="Height"
117 Weight="Weight"
118 family_history_with_overweight="Family History with Overweight"
119 SCC="Calories Consumption Monitoring"
120 FAF="Physical Activity Frequency"
121 MTRANS="Transportation Used"
122 TUE="Time Using Technology Devices"
123 NObeyesdad="Obesity Result";
124
125 *merge two data sets*;
126 data merged_dataset;
127 set GROUP.TB1;
128 if _N_ = 1 then do;
129 set GROUP.TB2(keep=Gender Age Height Weight SCC FAF MTRANS TUE NObeyesdad);
130 end;
131 else do;
132 set GROUP.TB2(keep=Gender Age Height Weight SCC FAF MTRANS TUE NObeyesdad) point=_N_;
133 end;
134 run;
135
136 proc contents data=merged_dataset order=varnum;
137 run;
138
139 *descriptive statistics*;
140 proc means data=merged_dataset n mean std median min max;
141 vars age height weight fcvc ncp ch2o faf tue;
142 run;
143
144 *histogram of obesity level*;
145 proc sgplot data=merged_dataset;
146 vbar NObeyesdad / stat=freq datalabel;
147 /* yaxis label="Number"; */
148 xaxis label="Obesity level";
149 title "Number of Individuals in Each Obesity Level";
150 run;
151
152 *pie chart*;
153 data with_history;
154 set merged_dataset;
155 where family_history_with_overweight = 'yes';
156 run;
157
158 proc gchart data=with_history;
159 pie NObeyesdad;
160 title 'Pie plot of Obesity Level with Family History of Overweight';
161 run;
162
163 data without_history;
164 set merged_dataset;

```

```

164     where family_history_with_overweight = 'no';
165     run;
166
167 proc gchart data=without_history;
168     pie NObeyesdad;
169     title 'Pie plot of Obesity Level without Family History of Overweight';
170 run;
171
172
173
174 *Chi-Square Analysis*;
175 PROC LOGISTIC DATA=merged_dataset;
176 CLASS NObeyesdad;
177 CLASS FAVC CAEC SMOKE CALC;
178 MODEL NObeyesdad (ref="Normal_Weight") = FAVC FCVC NCP CAEC SMOKE CH2O CALC/ LINK=glogit stb rsquare;
179 RUN;
180 PROC LOGISTIC DATA=merged_dataset;
181 CLASS NObeyesdad;
182 CLASS SCC MTRANS;
183 MODEL NObeyesdad(ref="Normal_Weight") = SCC FAF MTRANS TUE/ LINK=glogit stb rsquare;
184 RUN;
185
186 *Random Forest*;
187 data eating_habit;
188     set merged_dataset;
189     if FAVC = 'yes' then FAVC = 1;
190     else FAVC = 0;
191
192     if SMOKE = 'yes' then SMOKE = 1;
193     else SMOKE = 0;
194
195     if CAEC = 'Always' then CAEC = 3;
196     else if CAEC = 'Frequently' then CAEC = 2;
197     else if CAEC = 'Sometimes' then CAEC = 1;
198     else if CAEC = 'no' then CAEC = 0;
199
200     if CALC = 'Always' then CALC = 3;
201     else if CALC = 'Frequently' then CALC = 2;
202     else if CALC = 'Sometimes' then CALC = 1;
203     else if CALC = 'no' then CALC = 0;
204
205     if NObeyesdad = "Insufficient_Weight" then NObeyesdad = 1;
206     else if NObeyesdad = "Normal_Weight" then NObeyesdad = 2;
207     else if NObeyesdad = "Obesity_Type_I" then NObeyesdad = 3;
208     else if NObeyesdad = "Obesity_Type_II" then NObeyesdad = 4;
209     else if NObeyesdad = "Obesity_Type_III" then NObeyesdad = 5;
210     else if NObeyesdad = "Overweight_Level_I" then NObeyesdad = 6;
211     else if NObeyesdad = "Overweight_Level_II" then NObeyesdad = 7;
212     run;
213
214 proc hpforest data=eating_habit MAXTREES=250 MAXDEPTH=20;
215     target NObeyesdad;
216     input FAVC FCVC NCP CAEC SMOKE CH2O CALC;
217 run;
218
219
220
221 data lifestyle_habit;
222     set merged_dataset;
223     if SCC = 'yes' then SCC = 1;
224     else SCC = 0;
225
226     if MTRANS = 'Automobile' then MTRANS = 1;
227     else if MTRANS = 'Bike' then MTRANS = 3;
228     else if MTRANS = 'Motorbike' then MTRANS = 2;
229     else if MTRANS = 'Public_Transportation' then MTRANS = 4;
230     else if MTRANS = 'Walking' then MTRANS = 5;
231
232     if NObeyesdad = "Insufficient_Weight" then NObeyesdad = 1;
233     else if NObeyesdad = "Normal_Weight" then NObeyesdad = 2;
234     else if NObeyesdad = "Obesity_Type_I" then NObeyesdad = 3;
235     else if NObeyesdad = "Obesity_Type_II" then NObeyesdad = 4;
236     else if NObeyesdad = "Obesity_Type_III" then NObeyesdad = 5;
237     else if NObeyesdad = "Overweight_Level_I" then NObeyesdad = 6;
238     else if NObeyesdad = "Overweight_Level_II" then NObeyesdad = 7;
239
240     run;
241
242
243 proc hpforest data=lifestyle_habit MAXTREES=250 MAXDEPTH=20;
244     target NObeyesdad;
245     input SCC FAF MTRANS TUE;
246 run;

```


246
247
248
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251
252
253

The CONTENTS Procedure

Data Set Name	WORK.MERGED_DATASET	Observations	2111
Member Type	DATA	Variables	17
Engine	V9	Indexes	0
Created	11/02/2023 13:37:29	Observation Length	152
Last Modified	11/02/2023 13:37:29	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	3
First Data Page	1
Max Obs per Page	861
Obs in First Data Page	832
Number of Data Set Repairs	0
Filename	/saswork/SAS_work6DF300002AEF_odaws02-usw2.oda.sas.com/SAS_work964900002AEF_odaws02-usw2.oda.sas.com/merged_dataset.sas7bdat
Release Created	9.0401M7
Host Created	Linux
Inode Number	536895845
Access Permission	rw-r--r--
Owner Name	u63615521
File Size	512KB
File Size (bytes)	524288

Variables in Creation Order

#	Variable	Type	Len	Format	Informat	Label
1	Gender	Char	6	\$6.	\$6.	Gender
2	Age	Num	8	BEST12.	BEST12.	Age
3	Height	Num	8	BEST12.	BEST12.	Height
4	Weight	Num	8	BEST12.	BEST12.	Weight
5	family_history_with_overweight	Char	3	\$3.	\$3.	Family History with Overweight
6	FAVC	Char	5	\$5.	\$5.	Frequent Consumption of High Caloric Food
7	FCVC	Num	8	BEST12.	BEST12.	Frequency of Consumption of Vegetables
8	NCP	Num	8	BEST12.	BEST12.	Number of Main Meals
9	CAEC	Char	12	\$12.	\$12.	Consumption of Food between Meals
10	SMOKE	Char	5	\$5.	\$5.	Smoking Status
11	CH2O	Num	8	BEST12.	BEST12.	Consumption of Water Daily
12	CALC	Char	12	\$12.	\$12.	Consumption of Alcohol
13	NObeyesdad	Char	19	\$19.	\$19.	Obesity Result
14	SCC	Char	3	\$3.	\$3.	Calories Consumption Monitoring

Variables in Creation Order						
#	Variable	Type	Len	Format	Informat	Label
15	FAF	Num	8	BEST12.	BEST12.	Physical Activity Frequency
16	MTRANS	Char	21	\$21.	\$21.	Transportation Used
17	TUE	Num	8	BEST12.	BEST12.	Time Using Technology Devices